Team Control Number

16895

Problem Chosen

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2025

MidMCM

Summary Sheet

Designing a Modern Zoo: Planning and Optimizing Animal Exhibits and Visitor Experience through Mathematical Modeling

Summary

This project designs a 1-square-kilometer wildlife-friendly zoo following the missions of modern zoos: conservation, education, research, and recreation. The goal is to create a scientific and practical layout that serves animals, visitors, and employees while following real-world standards.

To select animal species, we used two key factors: **the IUCN conservation value and the species' average captive lifespan**. We built **a priority-score** model that combines these two factors with assigned weights. Conservation value was given a higher weight because protecting endangered species is a major mission of wildlife-focused zoos. Using reliable data and mathematical functions, we calculated priority scores for 36 representative species. Species with the highest scores—such as orangutans, rhinos, elephants, and certain reptiles—were chosen for the final zoo design.

Next, we used the China Zoo Design Standard to determine minimum exhibit areas, land-use proportions, moat widths, and functional-zone distribution. The 1 km² zoo was divided into exhibit zones, an education center, walking paths, and service areas. Visitor-experience modeling was also included: using distance-satisfaction analysis, we placed routes to keep segment walking distances near 500 meters, which produces high satisfaction.

The results show a complete and realistic zoo plan that meets all four missions. The exhibits fit within land-use limits; visitor routes are efficient and comfortable; and animals receive proper space and safety. The model balances scientific reasoning with practical design.

Overall, our work demonstrates how mathematical modeling and real biological data can be combined to guide zoo planning. The final design provides a strong foundation for building a modern zoo that protects wildlife while offering meaningful experiences for people.

Keywords: species selection, conservation value, lifespan data, exhibit design, visitor routes, zoo layout, mathematical modeling

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1 Introduction

1.1 Problem background

In today's world, biodiversity is facing unprecedented challenges. Habitat loss, climate change, and human disturbances have placed many species in a highly vulnerable position. Modern zoos are no longer merely places for viewing animals; instead, they have become integrated platforms for wildlife conservation, scientific research, education, and recreation.

This project, commissioned by the **Council Obsessed with Maximizing Animal Prosperity (*COMAP*)**, aims to design a 1-square-kilometer wildlife-friendly zoo. The goal of the park is not only to provide animals with a safe and suitable habitat, but also to support scientific research, educational activities, and enjoyable visitor experiences. The problem requires us to consider the needs of three groups—animals, employees and visitors—and to optimize layout, facility placement, and visitor routes accordingly.

Specifically, the animal section must ensure animal health and activity; the employees section must ensure safety and work efficiency; and the visitor section must provide a rich, meaningful, and safe experience. Existing research indicates that high-quality modern zoos, which operate under the four missions of "conservation + education + research + recreation" perform significantly better than traditional zoos that function only as exhibition spaces.

Through the design of this project, we aim not only to propose a mathematically rigorous optimization model, but also to present a responsible and engaging paradigm for zoo planning in terms of spatial layout and visitor experience. By balancing animal welfare, staff safety, and visitor learning, our design seeks to integrate both conservation and entertainment into a unified and meaningful approach.

1.2 Restatement of problem

1.2.1 Four Major Objectives: Conservation, Research, Education, and Recreation

The mission of modern zoos is commonly summarized into four core objectives: wildlife conservation, scientific research, public education, and recreation:

Conservation: To provide animals with an environment suitable for living, breeding, and restoring endangered populations.

Scientific Research: To support biological and ecological studies by offering observation spaces, behavioral research opportunities, and reliable platforms for data collection related to animal welfare and ecosystem science.

Education: To communicate knowledge of wildlife protection, biodiversity, and ecological systems to the public—especially students—so that visitors not only observe animals but also gain meaningful learning experiences.

Recreation: To ensure a safe, peasant, and engaging visit for guests. Recreational elements increase visitor satisfaction and attendance, while also supporting the broader goals of education and conservation.

1.2.2 Three Categories of Stakeholders: Animals, Employees, and Visitors

Animals: The design must account for animal safety, welfare, adequate activity space, population health, and the potential impact of interactions with visitors and staff.

Employees: Zoo staff—such as keepers, veterinarians, and management personnel—require a secure working environment, efficient facility access, and a reasonable layout that supports daily operations.

Visitors: Visitors' needs include safety (avoiding dangerous encounters and minimizing cross-interference), experience quality (clear pathways, interactive elements, and educational information), and convenience (rest areas, drinking water, restrooms, and other amenities).

The problem requires balancing the needs of all three groups within the layout design. For example, staff pathways should not conflict with visitor routes, and animal exhibits must allow visibility for visitors while maintaining appropriate buffers and safety barriers.

2 Assumptions

To make our model and later planning easier, we make several assumptions about the zoo design. All details and numbers in these assumptions follow the *China Zoo Design Standard (CJJ267-2017)*[1].

(1) According to the standard, a zoo with more than 50 hectares of suitable land (land with a ground slope under 25%) is a large zoo. Our zoo area is 1 square kilometer, which is 100 hectares, and we assume all of it is suitable land. So our zoo is considered a large zoo and can hold more than 120 animals or species.

(2) For a large zoo, the standard says that the building areas should follow these limits:

- animal exhibit buildings ≤ 6.5%
- animal support facilities ≤ 1.5%
- education buildings ≤ 0.7%
- service and recreation buildings ≤ 2.9%
- management buildings ≤ 1.4%

In our 100-hectare zoo, these areas must be smaller than:6.5 ha, 1.5 ha, 0.7 ha, 2.9 ha, and 1.4 ha.

The standard also says:

• roads and paved areas ≤ 17% (≤ 17 ha in our zoo)

• green space ≥ 70% (≥ 70 ha in our zoo)

(3) The standard gives a formula for visitor capacity. Using the idea of "maximizing animal protection land" and "giving visitors the widest hard-surface area," we estimate that our zoo can hold about **15,666 visitors** at one time.

(4) There are two types of roads: visitor roads and staff-only management roads. According to road-width rules:

• Main roads for service vehicles or sightseeing carts should be 5.0–7.0 m wide. We assume **7.0 m** for our main roads.

• Staff-only management roads should be 4.0–6.0 m wide. We assume **6.0 m**.

• Walking paths for visitors follow the secondary standard of 3.5–5.0 m. We choose **5.0 m** for our walking paths.

(5) Barrier Design Between Animals and Visitors. The minimum barrier width between animals and visitors should not be less than 1.5 meters. Because some animals have long limbs (like an elephant's trunk), we simplify this rule and use a uniform barrier width of 2 meters in our model for animals without special safety considerations. For barriers between neighboring animal enclosures, we also follow this rule for safety and simplicity.

(6) Building Area Calculations. According to the formulas in the design standard, we assume the area of the animal exhibition buildings includes both the indoor enclosure buildings and the animal support facility buildings:

Animal exhibition + support facilities area:

$1.05 \times 0.065 \times 100 + 1.05 \times 0.015 \times 100$

$= 6.825 + 1.575$

$= 8.4$ hectares $= 0.084$ km²

Animal education building area:

$1.10 \times 0.007 \times 100$

$= 0.77$ hectares $= 0.0077$ km²

Service and recreation building area:

$1.28 \times 0.029 \times 100$

$= 3.712$ hectares $= 0.03712$ km²

(This area is mainly for visitor-use spaces.)

All above detailed calculations follow the China Zoo Design Standard (CJJ267-2017).

(7) Types of Animals in Exhibition Areas. According to the standard, the zoo animals are divided into seven categories:

• Primates (such as chimpanzees, gibbons, baboons, macaques)

• Felidae / Cat species (such as lions, tigers, cheetahs)

• Canidae / Dog species (such as wolves, foxes)

• Large herbivores (such as elephants, rhinos, hippos, giraffes)

• Bears (black bears, brown bears, polar bears)

• Birds

• Reptiles (such as turtles, snakes, lizards)

(7) Buffer Zone Width for Each Exhibit

The buffer zones (such as safety moats or trenches) required for different animal groups are shown in the table below. For each group, we use the maximum width listed in the design standard to ensure safety.

Table1: Required Buffer Widths for Animal Exhibits

Primates

Species	Moat Width (m)
Chimpanzee	4.0
Orangutan	4.0
Gibbon	4.0
Baboon	4.0
Macaque	4.0
Maximum Used in Model	**4.0**

Felidae (Big Cats)

Species	Moat Width (m)
African Lion	7.5
Tiger	7.5
Cheetah	6.5
Maximum Used in Model	**7.5**

Large Herbivores

Species	Moat Width (m)
Elephant	3.5
Rhino	3.0
Hippo	3.0
Giraffe	4.0
Maximum Used in Model	**4.0**

Bears

Species	Moat Width (m)
Black Bear	3.5
Brown Bear	4.5
Polar Bear	4.0

Species	Moat Width (m)
Maximum Used in Model	**4.5**

3 Presentation of Model and Solution

3.1 Animal Species Selection

Based on the seven animal categories listed in the *China Zoo Design Standard (CJJ267-2017)* and their typical species, we looked up the IUCN Red List of Threatened Species [2]to find the conservation status (endangerment level) of each animal. We then assigned a score to each conservation level so we can use it in our model. Figure 1 shows the nine IUCN conservation levels, and Table 2 gives the scoring rules we use for each level.

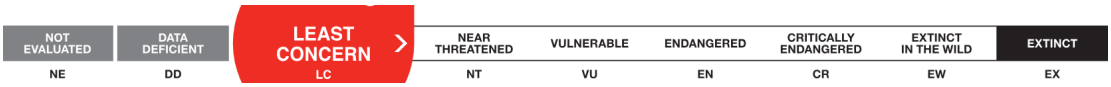


Figure 1. Classification of Animal Conservation Levels

Table 2. Scoring Rules for IUCN Threat Levels

Threat Level	Name	Score
LC	Least Concern	2
NT	Near Threatened	4
VU	Vulnerable	6
EN	Endangered	8
CR	Critically Endangered	10

To decide which species to include in our zoo, we consider two main factors:

1. **Conservation value** — This depends on the species' threat level. Since protecting wildlife is one of the most important missions of a modern zoo, conservation status becomes a key factor.
2. **Captive lifespan** — The lifespan of an animal affects the long-term sustainability of zoo operations. A longer-living animal has a lower replacement rate, allows visitors to form long-term connections, and improves zoo stability.

Both factors are important and cannot replace each other. Therefore, we construct a **species priority function**:

$$PS_i = (CV_i)^{w_1} \times (Lifespan_i)^{w_2}$$

where:

- PS_i= priority score of species i
- CV_i = conservation value score based on the IUCN category

- $Lifespan_i$= average captive lifespan of the species
- w_1, w_2 = weights for conservation value and lifespan

We use a **multiplicative model** instead of an additive model because if CV_i is low, the value of PS_i becomes much lower. This matches the idea that species with low conservation value should not be ranked highly in a conservation-focused zoo.

The values of CV_i come from the **IUCN Red List**, and the values of $Lifespan_i$ come from the dataset in *"Life Spans of Mammals, Birds, Amphibians, Reptiles, and Fish"* [3]Because conservation is the main mission of a wildlife-friendly zoo, we set the weights as: w_1=0.7, w_2=0.3.

Using the seven zoo animal categories from the *China Zoo Design Standard* and the data described above, we calculate the priority scores for each species and obtain the following table.

Table 3. Representative Animal Species and Their Priority Scores

ID	Species	IUCN Category	IUCN Score	Captive Lifespan	Priority Score
1	Chimpanzee	EN	8	56	7.63255821
2	Bornean Orangutan	CR	10	55	8.33930141
3	Gibbon	CR	10	30	7.78606345
4	Hainan Baboon	LC	2	27	4.31238017
5	Japanese Macaque	LC	2	29.3	4.37911585
6	African Lion	VU	6	30	6.27933520
7	Tiger	EN	8	25	6.91362165
8	Leopard	LC	2	20	4.08096084
9	Lynx	LC	2	26.8	4.30639156
10	Leopard Cat	LC	2	15	3.87784817
11	Bobcat	LC	2	32.3	4.46086090
12	Grey Wolf	LC	2	20	4.08096084
13	Red Fox	LC	2	15	3.87784817
14	Coyote	LC	2	20	4.08096084
15	Golden Jackal	LC	2	14	3.83168814
16	Bat-eared Fox	LC	2	13.8	3.82218110
17	Asian Elephant	EN	8	76	7.95352959
18	Rhinoceros	CR	10	45	8.14489657
19	Hippopotamus	VU	6	49	6.71923994
20	Giraffe	VU	6	28	6.22250553

ID	Species	IUCN Category	IUCN Score	Captive Lifespan	Priority Score
21	Black Bear	LC	2	33	4.47916343
22	Brown Bear	LC	2	47	4.79866876
23	Polar Bear	VU	6	38.2	6.48790888
24	Small Carnivorous Birds (e.g., Sparrowhawk)	LC	2	15.5	3.90012357
25	Medium-small Carnivorous Birds (e.g., Buzzard)	LC	2	25.3	4.26044865
26	Medium Raptors (e.g., Northern Goshawk)	LC	2	15	3.87784817
27	Large Raptors (e.g., Golden Eagle)	LC	2	22.8	4.17944331
28	Turtles <10 cm (e.g., Atlantic Ridley)	CR	10	3.6	6.48042914
29	Turtles >10 cm (e.g., Loggerhead Turtle)	VU	6	33	6.35980272
30	Land Snakes <60 cm (e.g., African Rufous Snake)	LC	2	4.3	3.17346598
31	Land Snakes <1.2 m (e.g., Yellow-red Rat Snake)	LC	2	7.8	3.47645106
32	Land Snakes <2.5 m (e.g., Corn Snake)	LC	2	21.8	4.14529642
33	Tree-dwelling Snakes <1.2 m (e.g., Golden Treesnake)	LC	2	4.3	3.17346598
34	Snakes >2.5 m (e.g., Python)	LC	2	14	3.83168814
35	Iguana	EN	8	2.5	5.60347605
36	Komodo Dragon	EN	8	8.9	6.21380674

Based on the data in Table 3, we plot a 3D graph showing the relationship among PS_i, CV_i, and $Lifespan_i$. The figure is shown below.

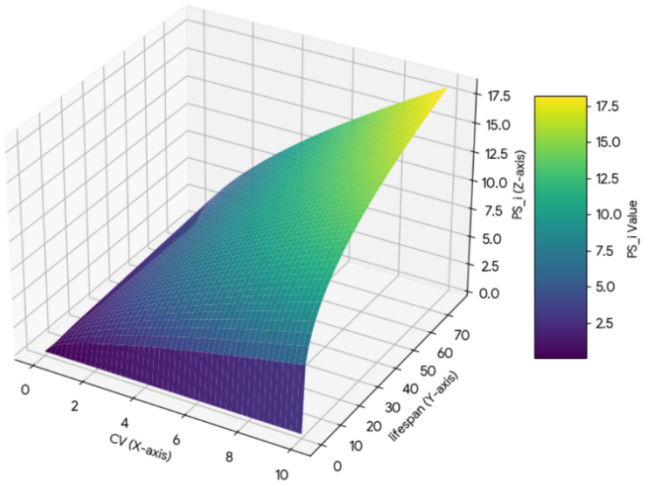


Figure2. 3D Plot of Animal Priority Scores

The figure shows that as the IUCN score (endangerment level) increases and the captive lifespan becomes longer, the priority score of the animal rises very quickly. This demonstrates that our model is scientifically reasonable in selecting species that are both important for conservation and suitable for long-term care in a zoo.

In addition, by ranking all animals according to their priority scores, we obtain Table 4**.**

Table 4. Priority Ranking of Animal Species for Conservation

Rank	ID	Species	IUCN Category	IUCN Score	Priority Score
1	2	Bornean Orangutan	CR	10	8.33930141
2	18	Rhinoceros	CR	10	8.14489657
3	17	Asian Elephant	EN	8	7.95352959
4	3	Gibbon	CR	10	7.78606345
5	1	Chimpanzee	EN	8	7.63255821
6	7	Tiger	EN	8	6.91362165
7	19	Hippopotamus	VU	6	6.71923994
8	23	Polar Bear	VU	6	6.48790888
9	28	Small Turtles (e.g., Atlantic Ridley Turtle, <10 cm)	CR	10	6.48042914
10	29	Large Turtles (e.g., Loggerhead Turtle, >10 cm)	VU	6	6.35980272
11	6	African Lion	VU	6	6.2793352
12	20	Giraffe	VU	6	6.22250553
13	36	Komodo Dragon	EN	8	6.21380674
14	35	Iguana	EN	8	5.60347605

Rank	ID	Species	IUCN Category	IUCN Score	Priority Score
15	22	Brown Bear	LC	2	4.79866876
16	21	Black Bear	LC	2	4.47916343
17	11	Bobcat	LC	2	4.4608609
18	5	Japanese Macaque	LC	2	4.37911585
19	4	Hainan Baboon	LC	2	4.31238017
20	9	Lynx	LC	2	4.30639156
21	25	Medium-sized Carnivorous Birds (e.g., Buzzard)	LC	2	4.26044865
22	27	Large Raptors (e.g., Golden Eagle)	LC	2	4.17944331
23	32	Terrestrial Snakes (<2.5 m, e.g., Corn Snake)	LC	2	4.14529642
24	8	Leopard	LC	2	4.08096084
25	12	Grey Wolf	LC	2	4.08096084
26	14	Coyote	LC	2	4.08096084
27	24	Small Carnivorous Birds (e.g., Sparrowhawk)	LC	2	3.90012357
28	10	Leopard Cat	LC	2	3.87784817
29	13	Red Fox	LC	2	3.87784817
30	26	Medium Raptors (e.g., Northern Goshawk)	LC	2	3.87784817
31	15	Golden Jackal	LC	2	3.83168814
32	34	Large Snakes (>2.5 m, e.g., Python)	LC	2	3.83168814
33	16	Bat-eared Fox	LC	2	3.8221811
34	31	Terrestrial Snakes (<1.2 m, e.g., Yellow-red Rat Snake)	LC	2	3.47645106
35	30	Terrestrial Snakes (<60 cm, e.g., African Rufous Snake)	LC	2	3.17346598
36	33	Arboreal Snakes (<1.2 m, e.g., Golden Treesnake)	LC	2	3.17346598

Based on the table above, LC represents species of Least Concern. To match the conservation goals of a wildlife-focused zoo, we remove all animals marked as LC.

Then, according to their priority scores, we select the top 14 species. The results are shown in Table 5.

Table 5. Selected Species for the Zoo

Primates	
ID	**Species**
1	Bornean Orangutan
4	Gibbon
5	Chimpanzee

Large Herbivores	
ID	**Species**
2	Rhinoceros
3	Asian Elephant
7	Hippopotamus
12	Giraffe

Felidae (Big Cats)	
ID	**Species**
6	Tiger
11	African Lion

Bears	
ID	**Species**
8	Polar Bear

Reptiles	
ID	**Species**
9	Small turtle (<10 cm), e.g., Atlantic ridley
10	Large turtle (>10 cm), e.g., Loggerhead turtle
13	Large monitor lizard, e.g., Komodo dragon
14	Iguana

From Table 5, we obtain the final list of animals for our zoo: primates, large herbivores, felines, bears, and reptiles — a total of five categories. According to the assumptions in Section 2, the indoor enclosure area for animals is 0.06825 km². Using the minimum required exhibit area for each species (from the *China Zoo Design*

Standard, CJJ267-2017) and the additional area needed when more animals are added, we plan the area for the five exhibit zones. The selected species and their exhibit areas are shown in Table 6.

Table 6. Zoo Animal Species and Exhibit Area Design

Primates

ID	Species	Minimum Exhibit Area (m²)
1	Orangutan	80 → 2000
4	Gibbon	60 → 2000
5	Chimpanzee	100 → 4000
Total Area (m²): 8000		
Requirement: moat width = 4 m		

Large Herbivores

ID	Species	Minimum Exhibit Area (m²)
2	Rhinoceros	500 → 4000
3	Elephant	900 → 10,000
7	Hippopotamus	350 → 2000
12	Giraffe	500 → 6000
Total Area (m²): 22,000		
Requirement: moat width = 4 m		

Felidae (Big Cats)

ID	Species	Minimum Exhibit Area (m²)
6	Tiger	200 → 15,000
11	African Lion	200 → 15,000
Total Area (m²): 30,000		
Requirement: moat width = 7.5 m		

Bears

ID	Species	Minimum Exhibit Area (m²)
8	Polar Bear	500 → 6000
Total Area (m²): 6000		
Requirement: moat width = 4 m		

Reptiles		
ID	**Species**	**Minimum Exhibit Area (m²)**
9	Small turtles (<10 cm) such as *Atlantic ridley*	200 →500
10	Larger turtles (>10 cm) such as *Loggerhead turtle*	200 →550
13	Komodo dragon	300 →600
14	Iguana	300 → 600
	Total Area (m²): 2250	

3.2 Visitor Experience and Spatial Layout Strategy

3.2.1 Visitor Experience Analysis

Using online information from zoos such as the Shanghai Zoo and Beijing Zoo, we build a simple function to describe the relationship between a visitor's walking distance and their satisfaction. We define the function as:

$$S = a + bL$$

where S is the visitor satisfaction level, and L is the walking distance.We use Excel to fit the data and create **Figure 3**.

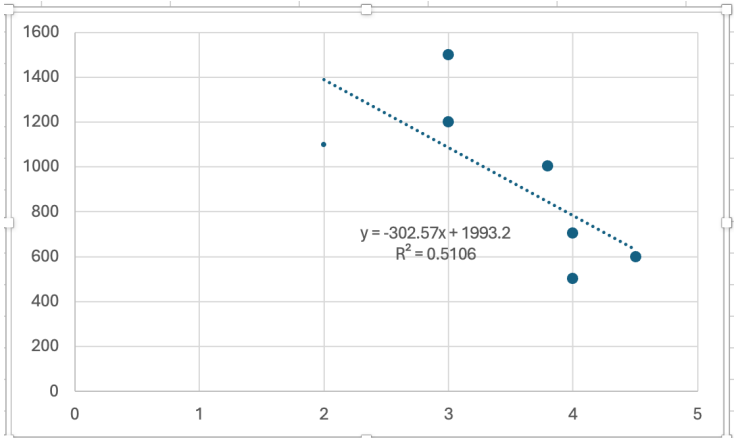


Figure 3. Walking Distance and Visitor Satisfaction

From the graph, we find that walking distance and satisfaction have a **negative relationship**. Based on zoo experience reports, the best walking distance for one segment in a zoo is **400–800 meters**, with the **highest satisfaction around 500 meters**. By placing activity areas and rest spots evenly along the route, we can improve the visitor experience.

3.2.2 Visitor Route Layout

The visitor route layout follows these main ideas:

(1) Avoid Repeated Paths. Visitors should be able to walk through the zoo **without turning back** or repeating sections. This lowers walking distance and makes the tour easier. The recommended route ensures that visitors see all major exhibits.

(2) Safe Distance Design. To keep visitors and animals safe, we use safety trenches outside some exhibit areas. The trench widths follow the standards in the *China Zoo Design Standard (CJJ267-2017)* (see Table 5 for details).

(3) Functional Area Placement. As stated in Section 2, the 1 km² zoo must include all functional zones.

- **Animal Exhibits (8.4 hectares)**: divided into five themed zones (e.g., Primate Zone, Big Cat Zone).
- **Education Center (0.77 hectares)**: placed near the entrance so visitors can easily reach information and rest areas.
- **Service and Recreation Buildings (3.712 hectares)**: spread evenly along the route instead of gathered in one place.

(4) **Rest and Service Nodes.**To reduce fatigue, the 3.712 hectares of service space are divided into many nodes, such as benches, small shelters, restrooms, vending machines, and small food stands. These nodes are placed along the main walking routes for maximum convenience.

In sum, through efficient routes and well-placed rest nodes, our layout combines:

- 8.4 hectares of exhibit areas
- 0.77 hectares of education space
- 3.712 hectares of service facilities

This creates a modern zoo environment with **shorter walking distances, easier rest access, and a more comfortable visitor experience**.

4 Analysis of Your Work

4.1 Implications of Our Zoo Design

Our work shows how a zoo can be planned in a way that is both scientific and responsible. First, our animal–selection model focuses on species that are **both endangered and able to live for a long time in captivity**. This means the zoo does not just show “popular” animals. Instead, it supports real wildlife conservation and gives scientists time to collect long-term data on these species.

Second, our layout design helps **visitors learn, explore, and enjoy the zoo** in a comfortable way. By planning walking distances of about 400–800 meters per segment and placing rest and service nodes along the route, visitors can see all five main animal zones without feeling too tired. Putting the education center in the core of the Zoo also helps students and families quickly reach information, exhibits, and activities that teach them about biodiversity and ecosystems.

Third, our model also considers **employees**, not only animals and visitors. Clear working routes, separated from visitor routes, and reasonable exhibit sizes make it easier for keepers, veterinarians, and other staff to feed animals, clean enclosures, and monitor health. This supports daily management and improves animal welfare.
Overall, our design helps the zoo meet its **four main goals**:

- **Conservation** – by choosing threatened species with high conservation value,
- **Research** – by supporting long-term studies on animal behavior and lifespan,
- **Education** – by creating an easy-to-understand layout and an education center,
- **Recreation** – by giving visitors smooth routes and comfortable places to rest.

In this way, our zoo becomes not only a place for fun, but also a place where animals, visitors, and employees all benefit from a well-planned system.

4.2 Sensitivity Analysis

Sensitivity analysis is used to test how much our results change when we adjust or remove an assumption.

Test 1: Changing the weight of conservation value

In our priority model, we set the weight of conservation value to 0.7 and lifespan to 0.3. We tested what happens if both weights are changed to 0.5 and 0.5.

Result:

Some long-living but less endangered animals became ranked higher. For example:

- Brown bear and black bear moved up several places.
- Rhinoceros and orangutan stayed high but slightly shifted.
- Iguana dropped because its lifespan is short.

This test shows that our original choice of weights really pushes the zoo toward **strong conservation**. If a future zoo wanted more focus on entertainment or stable "famous animals," it might choose a more balanced weighting like 0.5/0.5. Our work makes it clear how the **mission of the zoo** (protection vs. entertainment) directly changes which animals should be selected.

Test 2: Considering Protection Efficiency

If we calculate *conservation efficiency*, which means how much conservation value a species provides per unit of area, we define the function:

$$E_i = \frac{PS_i}{Area_{min_i}}$$

Here,

- E_i is the conservation efficiency of species i,
- PS_i is the species' priority score, and
- $Area_{min_i}$ is the minimum enclosure area required for that species.

Using the previously calculated values of PS_i, we compute the results as shown below.

Table 7. Ranking of Species by Conservation Efficiency

Rank	ID	Species	IUCN Category	IUCN Score	Minimum Enclosure Area (m²)	Conservation Efficiency
1	28	Turtle <10 cm (e.g., Atlantic ridley)	CR	10	0.16	40.5026821
2	30	Terrestrial snake <60 cm (African rufous snake)	LC	2	0.18	4.3
3	31	Terrestrial snake <1.2 m (Yellow-red rat snake)	LC	2	0.40	8.69112764
4	33	Arboreal snake <1.2 m (Golden treesnake)	LC	2	0.48	6.61138747
5	35	Iguana	EN	8	1	5.60347605
6	29	Turtle >10 cm (Loggerhead turtle)	VU	6	2	3.17990136
7	32	Terrestrial snake <2.5 m (Corn snake)	LC	2	1.5	2.76353095
8	34	Large snake >2.5 m (Python)	LC	2	2	1.91584407
9	36	Monitor lizard (e.g., Komodo dragon)	EN	8	4.5	1.38084594
10	24	Small raptor (e.g., Sparrowhawk)	LC	2	15	0.26000824
11	25	Medium-small raptor (Buzzard)	LC	2	21	0.20287851
12	26	Medium raptor (Northern goshawk)	LC	2	24	0.16157701
13	3	Gibbon	CR	10	60	0.12976772
14	2	Bornean orangutan	CR	10	80	0.10424127
15	16	Bat-eared fox	LC	2	40	0.09555453

Rank	ID	Species	IUCN Category	IUCN Score	Minimum Enclosure Area (m^2)	Conservation Efficiency
16	27	Large raptor (Golden eagle)	LC	2	50	0.08358887
17	1	Chimpanzee	EN	8	100	0.07632558
18	11	Bobcat	LC	2	60	0.07434768
19	5	Japanese macaque	LC	2	60	0.07298526
20	10	Leopard cat	LC	2	60	0.0646308
21	4	Baboon	LC	2	80	0.05390475
22	9	Lynx	LC	2	100	0.04306392
23	7	Tiger	EN	8	200	0.03456811
24	14	Coyote	LC	2	120	0.03400801
25	15	Golden jackal	LC	2	120	0.03193073
26	6	African lion	VU	6	200	0.03139668
27	12	Grey wolf	LC	2	150	0.02720641
28	13	Red fox	LC	2	150	0.02585232
29	8	Leopard	LC	2	200	0.0204048
30	19	Hippopotamus	VU	6	350	0.01919783
31	18	Rhinoceros	CR	10	500	0.01628979
32	21	Black bear	LC	2	300	0.01493054
33	23	Polar bear	VU	6	500	0.01297582
34	20	Giraffe	VU	6	500	0.01244501
35	22	Brown bear	LC	2	500	0.00959734
36	17	Asian elephant	EN	8	900	0.00883726

Result:

- Reptiles and small turtles rank extremely high because they require very little land but still provide conservation value.
- Large mammals rank very low because they take up a lot of land.

If the zoo's goal were **maximum conservation value per square meter**, the zoo would choose many reptiles and birds, not big mammals. However, for real zoos, conservation must be balanced with education, research, and recreation—where large

mammals play an important role. This test shows that **different goals lead to very different optimal species lists**.

Test 3: Changing the Visitor Walking-Distance Requirement

In our design, we assume that the best walking distance for one route segment is **400–800 meters**, with the most comfortable distance around **500 meters**. This is based on zoo reports and our fitted satisfaction curve.

Now we test what happens if the zoo wants to mainly serve **very young children or elderly visitors**, and changes the target segment length to **200–400 meters**.

Result:

To keep each segment shorter, the zoo would need:

- More small rest and service nodes,
- More path branches or shortcuts,
- Exhibits and service buildings placed closer together.

This would increase construction and maintenance costs, and some land might need to be moved away from animal enclosures to service and path areas.

This test shows that **visitor experience is very sensitive to the walking-distance assumption**. If the target group changes (for example, from teenagers to elderly visitors), the route design must be adjusted, and the balance between animal space and visitor space may also change. Our model gives a clear baseline for "normal" visitors, but future planners can adapt it for other groups by changing this assumption.

4.3 Strengths and Limitations

Our work has several important strengths. **First, our species-selection model is built on reliable scientific data.** We used the official IUCN conservation categories to measure how endangered each animal is, and we used published international studies to obtain accurate values for captive lifespans. We also followed the China Zoo Design Standard for key information such as animal categories, minimum exhibit areas, land-use proportions, and the size requirements for each functional zone. Because our inputs come from recognized scientific and national standards, our calculations match what a real zoo would need. **In addition, we created mathematical functions that describe the relationships among conservation value, lifespan, and priority score**. By graphing these functions, we were able to clearly see how different variables interact and how they affect species ranking. This helped us make decisions in a logical and evidence-based way.

Another strength of our work is that the final zoo design supports all four major missions of a modern zoo: conservation, education, research, and recreation. The species we selected have strong conservation importance, the exhibit sizes are realistic, and the layout allows visitors to follow an efficient route while also accessing learning spaces and resting areas. Our model also considers multiple stakeholders.

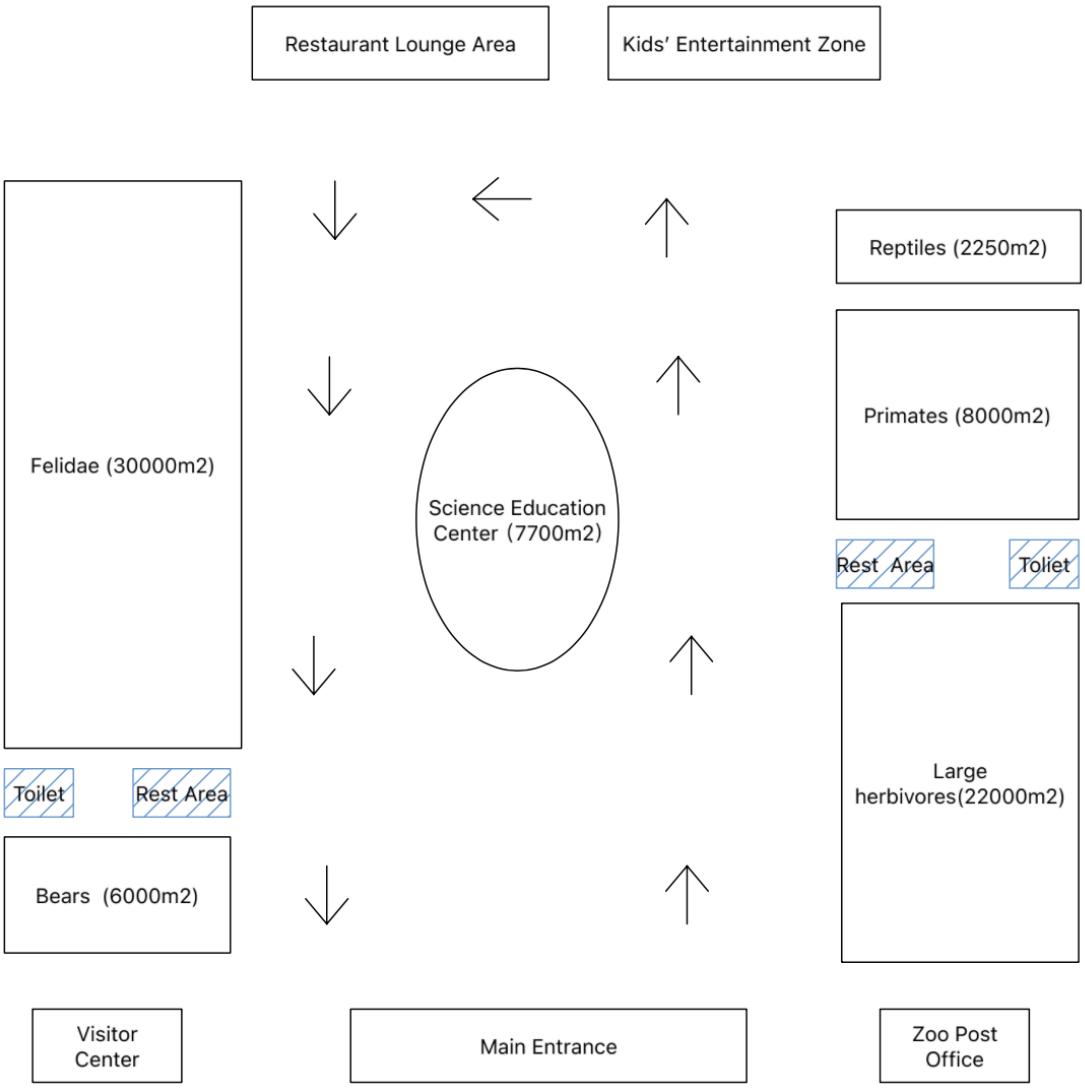
Animals receive enough space and proper care, visitors have a comfortable and enjoyable walking experience, and employees have clear work areas that help the zoo operate safely and smoothly. Together, these features show that our design is scientifically grounded and practical for real-world use.

However, our model also has several limitations. Although we used trusted sources, some important factors were simplified or not included. For example, we did not model animal behavior, diet needs, or the detailed enrichment spaces required to keep animals active and healthy. We used minimum exhibit areas from the standard, but real zoos often need larger or more complex environments, which could change space planning. Our visitor-satisfaction model is also simple because we did not have access to large real-world datasets; therefore, it may not fully represent how all visitor groups behave. Finally, our model does not simulate daily or yearly zoo operations, such as feeding costs, staff schedules, seasonal weather effects, or long-term changes in animal populations. These factors could affect how well the design works in the long run. Even with these limitations, our model provides a useful starting point for designing a zoo that balances scientific principles with practical planning.

5 Concluding Paragraph

In this project, we designed a 1-square-kilometer wildlife-friendly zoo that meets the needs of animals, visitors, and employees while supporting the four major missions of modern zoos: conservation, education, research, and recreation. By using reliable conservation data, scientific lifespan records, and national zoo-design standards, we built mathematical models that helped us select species, plan exhibit areas, and design visitor routes in a clear and logical way. Our results show that a balanced model can create a zoo layout that protects endangered animals, provides meaningful learning opportunities, and offers an enjoyable visiting experience. In the future, our work could be improved by adding more detailed data—such as feeding requirements, climate effects, or daily zoo operations—and by creating simulation models to study long-term changes in animal populations. Overall, our model provides a strong foundation for designing a modern zoo that is safe, educational, and sustainable, and it offers ideas that future researchers or planners can build upon.

6 Clearly Labeled Scale Drawing of Zoo Design



7 Visitor's Guide

Page 1 — Welcome to WildCare Zoo!

We are happy you are here! This guide will help you enjoy the zoo and learn as much as possible. If you follow our official walking route, you can see every animal without getting too tired. The path is a big loop that starts and ends at the Main Entrance.

1. Begin at the Herbivore Plains (Large Herbivores Area)

This is the best place to start because these animals live in the biggest spaces. You can see elephants, rhinos, giraffes, and hippos.

What to learn:
You can read signs about how these animals help their natural habitats and why many of them are endangered today.
For fun:
There are shaded platforms where you can take great photos and rest before moving on.
2. Then go to the Primate Forest (Primates Area)

Here you can meet orangutans, chimpanzees, and gibbons. They are very active and interesting to watch.
What to learn:
You can see how primates communicate, solve problems, and live in groups.
For fun:
Try the "Jump Like a Primate" activity to see how high you can jump compared to them!
3. Next, explore the Reptile Domain (Reptiles Area)

This area has turtles, lizards, and snakes. You can learn how reptiles survive in many different environments.
What to learn:
Look for the display that explains how temperature changes affect reptile behavior.
For fun:
There are small rest shelters along the path, so this is a good place to take breaks.

Page 2 — Continue Your Zoo Journey
4. Head to the Big Cat Territory (Felidae Area)

Here you will find lions, tigers, and other big cats. They are powerful animals that need large spaces.
What to learn:
You can read about how big cats hunt and why they are losing their habitats.
For fun:
Visit the tall viewing deck for the best look at the lions.
5. End your animal tour at the Bear Zone (Bear Area)

This area brings you back near the Main Entrance. You can learn about polar bears and how climate change affects them.
What to learn:
Read the boards about how warming weather impacts bears around the world.
For fun:
This place has a wide viewing area that makes it easy to take photos.
6. Stop at the Discovery Education Center (Center of the Zoo)

This building is in the middle of the zoo and is the best place to learn more.
Here you can find:
• Videos about conservation
• Hands-on models

• Species information screens
• Simple lessons about ecosystems
This is a great spot for school groups and curious families.
7. Rest at the Family Fun Zone & Dining Area (Back of the Zoo)

You will enjoy the food and rest area in the middle of the loop.
Here you will find:
• Restaurants and snack stands
• Shaded picnic tables
• Vending machines
• A small children's playground

It's the perfect place to relax.

We Hope You Enjoy Your Visit!

WildCare Zoo is designed to support four main goals: **animal protection, education, scientific research, and visitor recreation**.

By following this guide, you can explore the zoo comfortably and learn about the amazing animals we protect.

8 Memos

How Our Zoo Design Can Be Used in the Future

Our zoo was designed with one main purpose: to support the four most important missions of a modern zoo — **Conservation, Education, Research, and Recreation**. Everything in our plan, including the walking route, the habitat areas, and the location of the Education Center, helps us reach these goals.

1. Our Main Design Goals

• **Conservation:** We chose animals that need protection and gave them safe and comfortable habitats.
• **Education:** We placed the Education Center in the middle of the zoo so all visitors can learn easily.
• **Research:** The animals we selected help scientists study behavior, health, and long-term survival.
• **Recreation:** The walking path is simple, smooth, and enjoyable, so visitors have a good experience while learning.

2. How We Selected the Animals

We used two main ideas when choosing the animals:
• **How endangered the animal is** (using the official IUCN Red List)
• **How long the animal lives in captivity** (using trusted scientific studies)
Using these ideas, we chose 14 animals from five groups: **Primates, Large Herbivores, Big Cats, Bears, and Reptiles.**
These animals match the four missions and also fit well into the size of our zoo.

3. Can This Zoo Design Work for Other Animals?

Yes. Our design can easily be used with a different group of animals.

The reasons are:

• The **zones can be resized** or changed for new species.

• The **animal-selection method** (endangerment + lifespan) works for almost any animal.

• The **loop-shaped visitor route** is simple to adjust and can fit new habitats.

• The **four missions stay the same**, even if the animals change.

If another zoo wants more birds, more reptiles, or more small mammals, we can replace one zone with another without changing the whole plan.

4. Conclusion

Our zoo design is flexible, simple to update, and strongly focused on the four zoo missions. Even though we chose a specific set of animals for this project, the design can be adapted for many different animal collections in the future.

9 One-Page Letter

Dear COMAP Committee,

We are writing to share why we believe our zoo design is a great choice for your project. When planning our zoo, we focused on the four main purposes of modern zoos: **conservation, education, research, and recreation**. Our design was made to support all four goals in a clear and balanced way.

First, our zoo strongly supports **animal conservation**. We selected species based on two scientific ideas: **how endangered the animal is, and how long it can live in captivity**. This helped us choose animals that truly need protection and that can stay healthy in the zoo for many years. That is why our final list includes important endangered animals like orangutans, rhinoceroses, elephants, tigers, and polar bears.

Second, our design helps **education**. We placed the Education Center in the middle of the zoo so everyone can reach it easily. Along the walking path, visitors will find learning signs and displays that explain each animal's lifestyle and why the species is important. Students, families, and young children can all learn something new as they explore.

Third, our zoo supports **research**. Because we chose species with real conservation value and followed official zoo design standards, researchers will have safe, organized habitats where they can study animal behavior, health, and conservation needs.

Finally, our zoo gives visitors a great **recreational experience**. The visitor path is designed as a smooth loop with good walking distances, resting spots, and clear signs. Visitors start with big herbivores, then move to primates, reptiles, big cats, and finally

bears near the exit. This order keeps the experience fun and interesting. In the middle of the loop, people can relax in the dining area and children can play in the kids' zone.

In conclusion, our zoo design protects animals, teaches visitors, supports research, and makes the trip enjoyable for everyone. We believe our design fits COMAP's goals and would be a strong and responsible choice.

Thank you for reading our letter.

Sincerely,
Team 16895

Reference

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Report on Use of AI Tools

1.Baidu Fanyi, *Baidu Translate* (Nov. 17,2025 version)

Uploaded entire paper written in Mandarin to be translated into English.